CSC6033 - Week 6 Coding Project (P#6)
Turing Machine for Unary Addition

Irfan Ahmed

Unary Number Representation

In unary notation:

- A number n is represented by n occurrences of the symbol `0`.
- Two numbers are separated by a special symbol, e.g., `#`.

Examples:

- $2 \rightarrow$ `00`
- $3 \rightarrow$ `000`
- $2 + 3 \rightarrow$ `00#000` $\Rightarrow$ Result: `00000`

Objective

Design a Turing Machine (TM) that:

- Accepts two unary numbers separated by `#`
- Replaces `#` with `0` to merge the two sequences
- Replaces the last `0` with a blank (`B`) to mark the end
- Halts after completing the addition

Formal Definition of the Turing Machine

$\mathbf{M = (Q,\ \Sigma,\ \Gamma,\ \delta,\ q_0,\ B,\ F)}$

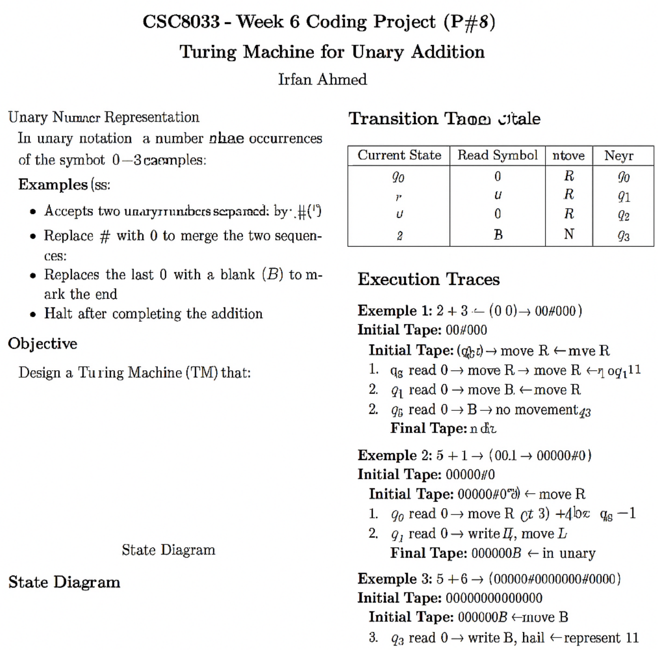
- $Q = \{q_0, q_1, q_2, q_3\}$ (Set of states)
- $\Sigma = \{0, \#\}$ (Input alphabet)
- $\Gamma = \{0, \#, B\}$ (Tape alphabet)

- δ = Transition function
- q_0 = Start state
- B = Blank symbol
- $F = \{q_3\}$ (Final state)

Transition Table

Current State	Read Symbol	Write Symbol	Move	Next State
q_0	0	0	R	q_0
q_0	#	0	R	q_1
q_1	0	0	R	q_1
q_1	B	B	L	q_2
q_2	0	B	N	q_3 (halt)

State Diagram



CSC8033 - Week 6 Coding Project (P#8)

Turing Machine for Unary Addition

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Unary Numncr Representation

In unary notation a number nhae occurrences of the symbot 0—3 caəmples:

Examples (ss:

- Accepts two unaryrunbers separacd: by .#(')
- Replace # with 0 to merge the two sequen-ces:
- Replaces the last 0 with a blank (B) to m-ark the end
- Halt after completing the addition

Objective

Design a Tu ring Machine (TM) that:

State Diagram

State Diagram

Transition Taoɑ ɔtale

Current State	Read Symbol	ntove	Neyr
q_0	0	R	q_0
r	u	R	q_1
u	0	R	q_2
2	B	N	q_3

Execution Traces

Exemple 1: 2 + 3 ← (0 0) → 00#000)
Initial Tape: 00#000
Initial Tape: (q) → move R ← mve R
1. q_8 read 0 → move R → move R ← oq$_1$11
2. q_1 read 0 → move B ← move R
2. q_5 read 0 → B → no movement q_3
Final Tape: n dz

Exemple 2: 5 + 1 → (00.1 → 00000#0)
Initial Tape: 00000#0
Initial Tape: 00000#0 ← move R
1. q_0 read 0 → move R (t 3) +4 bz q_8 −1
2. q_1 read 0 → write Π, move L
Final Tape: 000000B ← in unary

Exemple 3: 5 + 6 → (00000#0000000#0000)
Initial Tape: 00000000000000
Initial Tape: 000000B ← move B
3. q_3 read 0 → write B, hail ← represent 11

Figure 1: State Diagram of the Turing Machine for Unary Addition

Execution Traces

Example 1: $2 + 3 \to$ `00#000`

Initial Tape: `00#000`

Steps:

1. q_0, read 0 $\to$ move R
2. q_0, read 0 $\to$ move R
3. q_0, read $\#$ $\to$ write 0, move R $\to q_1$
4. q_1, read 0 $\to$ move R
5. q_1, read 0 $\to$ move R
6. q_1, read 0 $\to$ move R
7. q_1, read B $\to$ move L $\to q_2$
8. q_2, read 0 $\to$ write B, halt $\to q_3$

Final Tape: `000000B` $\Rightarrow$ represents 5 in unary

Example 2: $5 + 1 \to$ `00000#0`

Initial Tape: `00000#0`

Steps:

1. q_0, read 0 $\to$ move R $(\times 5)$
2. q_0, read $\#$ $\to$ write 0, move R $\to q_1$
3. q_1, read 0 $\to$ move R
4. q_1, read B $\to$ move L $\to q_2$
5. q_2, read 0 $\to$ write B, halt $\to q_3$

Final Tape: `000000B` $\Rightarrow$ represents 6 in unary

Example 3: $5 + 6 \to$ `00000#000000`

Initial Tape: `00000#000000`

Steps:

1. q_0, read 0 $\to$ move R $(\times 5)$
2. q_0, read $\#$ $\to$ write 0, move R $\to q_1$

3. q_1, read 0 → move R (×6)
4. q_1, read B → move L → q_2
5. q_2, read 0 → write B, halt → q_3

Final Tape: `00000000000B` ⇒ represents 11 in unary